Ethan Tizabgar

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EDUCATION

University of California, Irvine – Mechanical Engineering
B.S. Mechanical Engineering | Expected 2028
Cumulative GPA: 3.86/4.00

SKILLS

Technical Skills: SolidWorks (CSWA), Arduino, C++
Engineering: CAD Modeling, FEA, Technical Drafting, Prototyping, Mechanical Design
Coursework: Engineering Graphics & Design, Statics

WORK EXPERIENCE

SEPTEMBER 2025 - June 2026

Supplemental Instructor (Statics & Engineering Design) — *Santa Monica College*

- Provided supplemental instruction for statics and engineering design, emphasizing SolidWorks modeling, technical drafting, free-body diagrams, and equilibrium analysis.
- Developed practice problems, review materials, and exam preparation resources in collaboration with the instructor to address common conceptual challenges.
- Led structures review sessions and provided individualized support in problem-solving and SolidWorks modeling.

July 2022- January 2025

Swim Instructor — *Westside Aquatics*

- Instructed children ages 2–13 in swimming, adapting instruction to different developmental levels and learning needs.
- Developed individualized lessons while maintaining a safe learning environment and communicating student progress with parents.

ENGINEERING PROJECTS

October 2026 - Current

Dynamics Team Member — *UCI Anteater Hydrogen-Powered Vehicle (AHPV)*

- Collaborate with the dynamics team on steering system design and vehicle handling considerations for a hydrogen-powered endurance vehicle.
- Explore steering geometry, design constraints, and mechanical considerations to support vehicle stability and maneuverability.

May 2026 - June 2026

EMG-Controlled Robotic Hand — *Santa Monica College Team Project*

- Collaborated with a team on a tendon-driven robotic hand using servo actuation, springs, PTFE tendon routing, and 3D-printed components.
- Integrated a MyoWare 2.0 EMG muscle sensor with an Arduino Uno R4 WiFi to measure forearm muscle activity and translate contractions into gesture commands.
- Developed Arduino-based signal processing, calibration, and threshold-based classification to distinguish rock, paper, and scissors gestures.

January 2025 - February 2025

Wildfire-Response Rainwater System — *Santa Monica College Team Project*

- Designed a hilltop wildfire-response water tank and developed load-bearing support legs for uneven terrain, accounting for stability, weight distribution, and safety requirements.
- Refined SolidWorks models and prototypes based on testing results and instructor feedback.

July 2023 - August 2023

Go-Kart Design & Fabrication — *UCLA Engineering Summer Program*

- Led a team through the design, fabrication, and testing of a functional go-kart, including SolidWorks modeling, material selection, and assembly.
- Applied problem-solving and troubleshooting skills under pressure to address mechanical failures, optimize frame components, and improve overall vehicle performance.